Database Query Specification Report

1. Passenger & Payment Analytics

1. Full purchase history

Purpose: Displays complete ticket purchase and payment history per passenger.
SQL techniques used:

- INNER JOIN across 3 tables (Passenger → Ticket → Payment)
- Multi-table relationship traversal (1-to-many chains)
- ORDER BY for chronological sorting

2. Passengers with no purchased tickets

Purpose: Finds passengers who never made any ticket purchase.
SQL techniques used:

- LEFT JOIN (preserves all passengers)
- NULL filtering (anti-join pattern)
- WHERE clause for exclusion logic

3. Revenue by ticket type

Purpose: Calculates revenue and ticket volume per ticket category.
SQL techniques used:

- GROUP BY aggregation
- SUM() and COUNT() functions
- INNER JOIN (Ticket–Payment)

4. Revenue by payment method

Purpose: Analyzes revenue distribution across payment methods.
SQL techniques used:

- GROUP BY aggregation
- SUM() and COUNT()
- Direct aggregation on single table

5. Frequent passengers above average

Purpose: Identifies passengers with above-average ticket purchases.
SQL techniques used:

- GROUP BY aggregation
- Nested subquery for AVG calculation
- HAVING clause for filtering aggregated results
- Derived table (subquery in FROM)

6. Most recent payment per passenger

Purpose: Retrieves latest payment record for each passenger.
SQL techniques used:

- Correlated subquery
- MAX() aggregation function
- JOIN + filtering on computed scalar value

7. Ticket types purchased more than average

Purpose: Finds over-performing ticket types.
SQL techniques used:

- GROUP BY aggregation
- Nested subquery with AVG()
- HAVING clause filtering

8. Daily payment revenue

Purpose: Summarizes payments per day.
SQL techniques used:

- GROUP BY with DATE() function
- Aggregation (SUM, COUNT)
- Time-based grouping

9. Passengers using specific payment method

Purpose: Finds passengers who used a given payment method (Card).
SQL techniques used:

- EXISTS clause (semi-join)
- Correlated subquery
- Filtering based on relational condition

2. Trip & Schedule Analysis

10. Basic trip overview

Purpose: Combines trip and schedule timing information.
SQL techniques used:

- INNER JOIN between Trip and Schedule
- Relational mapping of scheduled vs actual data

11. Most used schedules

Purpose: Counts usage frequency of each schedule.
SQL techniques used:

- LEFT JOIN (preserves schedules with zero trips)
- GROUP BY aggregation
- COUNT() function

12. Trips by status

Purpose: Groups trips by operational status.
SQL techniques used:

- GROUP BY categorical aggregation
- COUNT(*) for frequency distribution

13. Delayed trips

Purpose: Detects trips departing later than scheduled.
SQL techniques used:

- INNER JOIN
- Comparison operators on time fields
- Filtering based on delay condition

14. Trips with no passengers

Purpose: Identifies unused or empty trips.
SQL techniques used:

- LEFT JOIN + NULL filtering (anti-join)
- GROUP BY + HAVING COUNT = 0 (second variant)
- Data integrity checking pattern

15. Passengers per trip

Purpose: Counts boardings per trip.
SQL techniques used:

- LEFT JOIN
- GROUP BY aggregation
- COUNT() function

16. Most popular trips

Purpose: Ranks trips by passenger usage.
SQL techniques used:

- GROUP BY aggregation
- ORDER BY DESC ranking logic
- COUNT() aggregation

17. High-performing trips

Purpose: Finds trips above average passenger volume.
SQL techniques used:

- Nested subquery with AVG()
- GROUP BY aggregation
- HAVING clause filtering
- Comparative analytics

18. Daily workload

Purpose: Measures trip activity per day.
SQL techniques used:

- GROUP BY date field
- COUNT aggregation

19. Busiest day

Purpose: Identifies day with highest trip count.
SQL techniques used:

- GROUP BY aggregation
- ORDER BY DESC

- LIMIT 1 (top-N query pattern)

20. Schedules generating most passengers

Purpose: Finds most productive schedules.
SQL techniques used:

- Multiple LEFT JOINs
- GROUP BY aggregation
- Multi-level relationship traversal

3. Route & Station Analysis

21. Route structure

Purpose: Displays full route layout with stations and timing.
SQL techniques used:

- INNER JOIN chain (Route → Route_stop → Station)
- Ordered hierarchical output

22. Most important stations

Purpose: Finds stations used in many routes.
SQL techniques used:

- GROUP BY aggregation
- COUNT(DISTINCT) for uniqueness
- JOIN for relational mapping
- ### 23. Complex routes

Purpose: Identifies routes with above-average stops.
SQL techniques used:

- GROUP BY aggregation
- Nested subquery for AVG comparison
- HAVING clause filtering

24. Station coverage percentage

Purpose: Measures station usage across all routes.
SQL techniques used:

- GROUP BY with COUNT(DISTINCT)

- CROSS JOIN for total reference value
- Percentage calculation

25. Most connected stations

Purpose: Finds highly connected transport hubs.
SQL techniques used:

- GROUP BY aggregation
- HAVING clause filtering
- COUNT(DISTINCT)

26. Route geographic breadth

Purpose: Measures route size by number of stops.
SQL techniques used:

- Scalar subquery
- COUNT aggregation
- Correlated reference to parent table

27. Zone density report

Purpose: Analyzes station distribution across zones.
SQL techniques used:

- GROUP BY categorical aggregation
- COUNT and COUNT(DISTINCT)

28. Station dwell time analysis

Purpose: Measures planned stop duration at stations.
SQL techniques used:

- JOIN operations
- Arithmetic on time intervals
- Filtering based on computed duration

29. Empty route integrity check

Purpose: Finds routes with no stops (data error detection).
SQL techniques used:

- LEFT JOIN anti-join pattern
- NULL filtering
- Data integrity validation

30. Station attribute validation

Purpose: Detects missing or invalid station data.
SQL techniques used:

- WHERE filtering
- NULL and empty string checks

4. Driver & Vehicle Management

31. Drivers above average workload

Purpose: Finds overloaded drivers.
SQL techniques used:

- CTE (WITH clause)
- GROUP BY aggregation
- Nested CTE for AVG comparison
- JOIN of derived tables

32. Vehicle utilization

Purpose: Measures total usage time per vehicle.
SQL techniques used:

- GROUP BY aggregation
- SUM of time intervals
- JOIN operation

33. Available drivers

Purpose: Finds currently free drivers.
SQL techniques used:

- NOT EXISTS anti-join
- Correlated subquery
- Time range filtering

34. Available vehicles

Purpose: Finds currently available vehicles.
SQL techniques used:

- NOT EXISTS clause
- Correlated subquery
- Time interval filtering

35. Full assignment details

Purpose: Shows complete driver-vehicle assignments.
SQL techniques used:

- Multi-table INNER JOIN
- Relationship mapping

36. Least used vehicles

Purpose: Identifies underutilized vehicles.
SQL techniques used:

- LEFT JOIN
- GROUP BY aggregation
- ORDER BY ASC ranking
- LIMIT clause

37. Most recent assignment

Purpose: Retrieves latest assignment record.
SQL techniques used:

- Scalar subquery with MAX()
- Filtering on computed value

38. Idle drivers

Purpose: Measures driver inactivity time.
SQL techniques used:

- LEFT JOIN
- GROUP BY aggregation
- Time difference computation
- MAX() aggregation

39. Daily workload

Purpose: Counts assignments per day.
SQL techniques used:

- GROUP BY date extraction
- COUNT aggregation
- Time-based analysis